

(a) Mean per-bin R^2

(b) RMSE

Raw-sequence models

CNN

CNN + RC augmentation

CNN + RC + strand averaging

CNN + RC + attention pooling

ConvNeXt

Nucleotide Transformer

NT: final layer

NT: layer 6

NT: learned layer mixture

NT: IA3 adaptation

Motif representation

Sparse motif model

0.3 0.4 0.5 0.6

Higher is better

Points: five-fold mean; intervals: ± 1 sample SD.

40 50 60 70

Lower is better

